QF1100 Introduction to Quantitative Finance — Final Exam Cheatsheet @stinkray77 AY2025/2026 Sem 2

1. Theory of Interest

Accumulation Functions

Simple: $a(t) = 1 + rt$ **Compound:** $a(t) = (1+r)^t$ [default]

Continuous: $a(t) = e^{\delta t}$

Accumulated value of principal P: $A(t) = P \cdot a(t)$

Frequency of Compounding

Nominal rate $r^{(p)}$ paid p times/yr:

$$a(t) = \left(1 + \tfrac{r^{(p)}}{p}\right)^{pt}$$

Effective annual rate:

$$r_e = \left(1 + \tfrac{r^{(p)}}{p}\right)^{p} - 1$$

Equivalence of two nominal rates:

$$\left(1 + \tfrac{r^{(p)}}{p}\right)^{p} = \left(1 + \tfrac{R^{(q)}}{q}\right)^{q}$$

$r_e > r^{(p)}$ for any $p > 1$. Continuous compounding maximises r_e for a given nominal rate.

Force of Interest

$$\delta(t) = \frac{a'(t)}{a(t)} = \frac{d}{dt}\ln a(t)$$

Constant δ: $a(t) = e^{\delta t}$, $\delta = \ln(1+r_e)$, $r_e = e^{\delta} - 1$.

Variable force of interest:

$$a(t) = \exp\Big(\int_0^t \delta(u)\,du\Big) \qquad a(s,t) = \exp\Big(\int_s^t \delta(u)\,du\Big)$$

Key Conversions (quick ref)

Convert	Formula
$r^{(p)} \to r_e$	$(1 + r^{(p)}/p)^p - 1$
$r_e \to r^{(p)}$	$p[(1+r_e)^{1/p} - 1]$
$r_e \to \delta$	$\ln(1+r_e)$
$\delta \to r_e$	$e^{\delta} - 1$
$r^{(p)} \to r^{(q)}$	equate $(1 + r^{(p)}/p)^p = (1 + r^{(q)}/q)^q$

2. Present Value & Cash Flows

Present Value (single payment)

$$\mathrm{PV} = \frac{c}{a(t)} = \frac{c}{(1+r)^t}$$

Discount factor: $v = \frac{1}{1+r}$, so $\mathrm{PV} = c \cdot v^t$

PV of Cash Flow Stream

$C = \{(c_i, t_i)\}_{i=1}^n$:

$$\mathrm{PV}(C) = \sum_{i=1}^n \frac{c_i}{(1+r)^{t_i}}$$

Time value at t: $\mathrm{TV}_t(C) = \mathrm{PV}(C) \cdot (1+r)^t$

Annuities

Let r = effective rate per period, A = payment per period, n = number of periods.

Ordinary annuity (payment at *end* of each period):

$$\mathrm{PV} = \frac{A}{r}\Big[1 - \frac{1}{(1+r)^n}\Big]$$

Annuity-due (payment at *start* of each period):

$$\mathrm{PV} = \frac{A(1+r)}{r}\Big[1 - \frac{1}{(1+r)^n}\Big]$$

$\Rightarrow$ PV(due) = PV(ordinary) $\times(1+r)$

Perpetuity (ordinary, payments at end):

$$\mathrm{PV} = \frac{A}{r}$$

Perpetuity-due (payments at start):

$$\mathrm{PV} = \frac{A(1+r)}{r}$$

Future value (ordinary annuity at $t = n$):

$$\mathrm{FV} = \frac{A}{r}[(1+r)^n - 1]$$

Deferred annuity (n payments, first at $t = m+1$):

$$\mathrm{PV} = \frac{A}{r}\Big[1 - \frac{1}{(1+r)^n}\Big] \cdot \frac{1}{(1+r)^m}$$

Discount ordinary annuity PV back by m periods.

Rate mismatch: Always convert nominal rate to *effective rate per payment period* before applying annuity formula. E.g. 6% p.a. convertible semi-annually $\to$ effective monthly rate $= (1+0.03)^{1/6} - 1$.

NPV & IRR Decision Rules

NPV criterion: Invest if NPV > 0. Prefer A over B if $\mathrm{NPV}(A) > \mathrm{NPV}(B)$.

IRR criterion: Invest if IRR $>$ required return. Prefer A over B if $\mathrm{IRR}(A) > \mathrm{IRR}(B)$.

IRR is r s.t. $\mathrm{PV}(C) = 0$. **Uniqueness:** if $c_0 < 0$, $c_k \ge 0\ \forall k \ge 1$, $\sum c_k > 0 \Rightarrow$ unique IRR.

Newton–Raphson: $r_{n+1} = r_n - f(r_n)/f'(r_n)$

Growing Annuity & Perpetuity

First payment A at $t = 1$, grows at rate g/period, n periods:

$$\mathrm{PV} = \frac{A}{r-g}\Big[1 - \Big(\frac{1+g}{1+r}\Big)^n\Big] \quad (r \ne g) \qquad r = g: \frac{nA}{1+r}$$

Growing perpetuity ($r > g$): $\mathrm{PV} = A/(r-g)$.

Loans, Refinancing & Last Payment

Loan L, n payments A, rate r/period: $L = \frac{A}{r}[1-(1+r)^{-n}]$, $A = \frac{Lr}{1-(1+r)^{-n}}$

Balance after m payments: $L_m^{\mathrm{bal}} = \frac{A}{r}[1-(1+r)^{-(n-m)}]$ (prospective)

$L_m^{\mathrm{bal}} = L(1+r)^m - \frac{A}{r}[(1+r)^m - 1]$ (retrospective)

Irregular last payment (n non-integer): make $\lfloor n \rfloor$ full payments of A plus final smaller payment B.

- B at $t = \lfloor n \rfloor$: set L = PV of $\lfloor n \rfloor - 1$ payments of A plus B at $t = \lfloor n \rfloor$.
- B at $t = \lfloor n \rfloor + 1$: use $L^{\mathrm{bal}}_{\lfloor n \rfloor}$, then $B = L^{\mathrm{bal}}_{\lfloor n \rfloor}(1+r)$.

3. Bonds

Notation

F	face value ($= R$ unless stated)
R	redemption value
c	nominal coupon rate (annual)
m	coupon payments per year
n	total coupon payments
λ	nominal bond yield (annual)
P	bond price

Bond Pricing Formula

$$P = \frac{R}{(1+\lambda/m)^n} + \sum_{i=1}^n \frac{cF/m}{(1+\lambda/m)^i}$$

$$= F\left[\frac{1}{(1+\lambda/m)^n} + \frac{c/m}{\lambda/m}\Big(1 - \frac{1}{(1+\lambda/m)^n}\Big)\right] \qquad (F = R)$$

Premium/discount form ($F = R$): $P = F + F \cdot \frac{c-\lambda}{\lambda}[1 - (1+\lambda/m)^{-n}]$

$c > \lambda \Rightarrow P > F$ (premium); $c = \lambda \Rightarrow P = F$ (par); $c < \lambda \Rightarrow P < F$ (discount).

Makeham formula (alternative; let $K = R/(1+\lambda/m)^n$):

$$P = K + \frac{c}{\lambda}(F - K) \qquad (F = R \text{ case})$$

Recursive price: $P_{k+1} = P_k(1+\lambda/m) - cF/m$ (price after k-th coupon).

Variable Coupon Bonds

Split into groups with constant coupon rate; price each group as a separate annuity + shared redemption PV.

Exam pattern (Q3 midterm): Bond with different coupon rates for each 5-year block and $R \ne F$. Price each block as an annuity discounted at yield λ, then add PV of redemption at maturity.

Determining Bond Yield (Newton–Raphson)

$f(\lambda) = P(\lambda) - P_{\mathrm{market}}$, iterate $\lambda_{n+1} = \lambda_n - f(\lambda_n)/f'(\lambda_n)$.

$$P(\lambda) = \frac{R}{(1+\lambda/m)^n} + \frac{cF/m}{\lambda/m}\Big[1 - \frac{1}{(1+\lambda/m)^n}\Big]$$

$f'(\lambda)$ for annual coupon bond ($m = 1$, $F = R$):

$$f'(\lambda) = -\sum_{i=1}^n \frac{i \cdot cF}{(1+\lambda)^{i+1}} - \frac{nF}{(1+\lambda)^{n+1}} = -\frac{D \cdot P(\lambda)}{1+\lambda}$$

where D = Macaulay duration. (In general: $f'(\lambda) = -\frac{D \cdot P}{m(1+\lambda/m)}$.)

Start: $\lambda_0 = c$. Price–yield: **inverse** & **convex** ($P \downarrow$ as $\lambda \uparrow$).

Macaulay Duration

$$D = \frac{\sum_{i=1}^n t_i \cdot \mathrm{PV}(c_i, t_i)}{\mathrm{PV}(C)} = \sum_{i=1}^n w_i t_i, \quad w_i = \frac{\mathrm{PV}(c_i)}{\mathrm{PV}(C)}$$

Special cases ($\mu = \lambda/m$, $\gamma = c/m$): Zero-coupon: $D = T$. Perpetuity : $D = (1+\lambda/m)/\lambda$.

Coupon bond, $F = R$ (general): $D = \dfrac{1+\mu}{m\mu} - \dfrac{1 + \mu + n(\gamma - \mu)}{m\gamma[(1+\mu)^n - 1] + m\mu}$ **(2)**

At par $P = F = R$ ($\mu = \gamma$): $D = \dfrac{1+\mu}{m\mu}\Big[1 - \dfrac{1}{(1+\mu)^n}\Big]$ **(3)**

$t_1 \le D \le t_n$; D independent of F and of yield compounding frequency.

4. Term Structure

Spot Rates & Forward Rates

Spot rate s_t: effective annual rate from 0 to t. Accumulation: $(1+s_t)^t$.

Forward rate $f_{j,k}$: rate from $t = j$ to $t = k$ (annual, same convention as spot).

No-arbitrage relation:

$$(1+s_k)^k = (1+s_j)^j (1+f_{j,k})^{k-j}$$

Note: $f_{0,t} = s_t$.

Continuous compounding version:

$$e^{s_k k} = e^{s_j j} \cdot e^{f_{j,k}(k-j)} \quad \Rightarrow \quad f_{j,k} = \frac{s_k k - s_j j}{k - j}$$

Bootstrapping Spot Rates

Use observed coupon bond prices to extract $s_1, s_2, \ldots$ sequentially:

For a coupon bond with annual payments:

$$P = \frac{c_1}{1+s_1} + \frac{c_2}{(1+s_2)^2} + \cdots + \frac{c_n + F}{(1+s_n)^n}$$

Solve for the unknown s_n given $s_1, \ldots, s_{n-1}$ already found.

Exam pattern: Extract s_1, s_2 from zero-coupon bond prices, then use coupon bond price to find s_3, then compute $f_{1,3}$.

Yield Curve Shapes

Normal (upward): economic expansion, low inflation.

Inverted (downward): pre-recession, high inflation.

Flat term structure: all $s_t = r$ equal. **Proof** (δ constant): continuous compounding $\Rightarrow a(t) = e^{rt}$, so $\delta(t) = a'(t)/a(t) = re^{rt}/e^{rt} = r$. Semiannual: $(1+s_t/2)^2 = e^{r_t}$; if s_t constant then r_t constant and vice versa. ✓

5. Asset Returns & Portfolios

Returns

Total return: $R = P_1/P_0$ **Rate of return:** $r = (P_1 - P_0)/P_0$ $R = 1 + r$

Short sell: $R_{\mathrm{short}} = P_1/P_0$ (same formula – negatives cancel).

Definitions & Identities

$\mathrm{Var}(X) = E[X^2] - E[X]^2$ $\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$ $\rho_{XY} = \dfrac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$

Expectation (linearity): $E(c) = c$; $E(cX) = cE(X)$; $E(X+c) = E(X) + c$; $E(aX+bY) = aE(X) + bE(Y)$

Variance: $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$; $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\mathrm{Cov}(X,Y)$

Covariance: $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$; $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$; $\mathrm{Cov}(aX, bY) = ab\,\mathrm{Cov}(X,Y)$

Bilinearity: $\mathrm{Cov}(aX+bY, Z) = a\,\mathrm{Cov}(X,Z) + b\,\mathrm{Cov}(Y,Z)$

Indep $\Rightarrow$ Cov $= 0$; Cov $= 0 \not\Rightarrow$ indep.

Asset Returns (notation)

Mean: $\mu_i = E(r_i)$; Var: σ_i^2; Cov: $\sigma_{ij} = \mathrm{Cov}(r_i, r_j)$

Short-sell: profit $= P_0 - P_1$, return $= -r_{\mathrm{long}}$, loss unlimited.

Portfolio of 2 Assets

Weights $w_1 = \alpha$, $w_2 = 1 - \alpha$ ($\alpha \in \mathbb{R}$ if short-selling allowed; $\alpha \in [0,1]$ otherwise).

$$\mu_p = \alpha\mu_1 + (1-\alpha)\mu_2$$

$$\sigma_p^2 = \alpha^2\sigma_1^2 + (1-\alpha)^2\sigma_2^2 + 2\alpha(1-\alpha)\rho_{12}\sigma_1\sigma_2$$

Global Minimum-Variance Portfolio (GMVP)

$$\alpha^* = \frac{\sigma_2^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$$

$$(\sigma_p^2)^* = \frac{\sigma_1^2\sigma_2^2(1-\rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$$

If $\rho = 1$: feasible set is a straight line. If $\rho = -1$: V-shape (zero variance achievable). If $-1 < \rho < 1$: curved bullet shape.

Efficient frontier: upper half of feasible set (including GMVP).

Constrained GMVP ($\mu_p \geq \bar{\mu}$)

If $\bar{\mu} > \mu_{\text{GMVP}}$, constraint is binding: set $\mu_p = \bar{\mu}$, solve $\alpha = \frac{\bar{\mu}-\mu_2}{\mu_1-\mu_2}$, substitute into σ_p^2.

No short-selling ($\alpha \in [0,1]$): if unconstrained $\alpha^* \notin [0,1]$, GMVP is the asset with lower σ.

Check: if $\alpha^* \notin [0,1]$ and short-selling not allowed, the GMVP is the asset with lower σ.

6. Forwards & Futures

Forward Contract

Obligation to buy/sell at forward price $F(t,T)$ at maturity T.

- Long payoff: $S_T - F(t,T)$
- Short payoff: $F(t,T) - S_T$
- Costs nothing to enter (no premium)

Forward Pricing (No-Arbitrage)

Non-dividend-paying asset:

$$F(0,T) = S_0 e^{rT} \text{ (cts)} \qquad F(0,T) = S_0(1+r)^T \text{ (disc)}$$

General t: $F(t,T) = S_t\, e^{r(T-t)}$

With storage cost (lump sum c at $t=0$):

$$F(0,T) = (S_0 + c)\, e^{rT}$$

With periodic storage (payments c at $t = 0, \Delta, 2\Delta, \ldots$): compute directly as annuity-due discounted to $t=0$.

Forward price does **not** depend on expected return μ of the underlying — only on S_0, r, T (and carry costs). This is a no-arbitrage result, not an expectation.

Arbitrage Construction

If $F(0,T) > S_0 e^{rT}$: short forward, buy asset (borrow S_0). Profit at T: $F - S_0 e^{rT} > 0$.

If $F(0,T) < S_0 e^{rT}$: long forward, short-sell asset (invest proceeds). Profit at T: $S_0 e^{rT} - F > 0$.

Futures vs. Forwards

	Forward	Futures
Settlement	At maturity	Daily MTM
Counterparty	Bilateral	Exchange

Daily P&L long futures: $\approx F_{t+1} - F_t$ (margin account).

Margin Account & Mark-to-Market

Setup: N contracts, contract size Q, futures price F_t.

Daily MTM: Gain/loss $= (F_t - F_{t-1}) \times Q \times N$ (positive for long, negative for short).

Margin call: If balance $<$ maintenance margin $\Rightarrow$ top up to *initial* margin. Top-up = initial − current balance.

Total contract value $= F_0 \times Q \times N$. Initial margin = fraction × TCV. Maintenance margin = fraction × TCV.

Top up to **initial**, not maintenance. Missing this costs full marks.

7. Hedging

Perfect Hedge (Forwards/Futures)

Enter opposite position to fix price and eliminate risk. Conditions for perfect futures hedge: (1) delivery date matches; (2) same underlying; (3) integer multiple of contract size; (4) sufficient liquidity.

Hedging Direction ($\rho > 0$)

Obligation	Risk	Action	P&L
Buy asset A	Price ↑	Long Futures A/B	$+(F_T - F_0)$
Sell asset A	Price ↓	Short Futures A/B	$+(F_0 - F_T)$

If $\rho_{A,B} < 0$: reverse direction.

Minimum-Variance Hedge

Hedge W units of asset A using futures on asset B (correlated). Short h futures contracts.

Cash flow at T: $y = -WS_T + (F_T - F_0)h$

$$\text{Var}(y) = W^2\text{Var}(S_T) - 2W\text{Cov}(S_T, F_T)h + \text{Var}(F_T)h^2$$

Optimal hedge ratio:

$$h^* = W \cdot \frac{\text{Cov}(S_T, F_T)}{\text{Var}(F_T)} = W \cdot \frac{\rho_{S,F}\,\sigma_S}{\sigma_F}$$

Minimum variance:

$$\text{Var}^*(y) = W^2\text{Var}(S_T)(1 - \rho_{S,F}^2)$$

Hedge effectiveness: $\rho_{S,F}^2$. Factor $\sqrt{1-\rho^2}$ measures residual risk.

h^* is in *units*, not contracts. Number of contracts $= h^*/(\text{contract size})$. Round to integer for real problems.

8. Options

Key Definitions

Call: right to *buy* at strike K. **Put**: right to *sell* at strike K.

Long = holder (pays premium); Short = writer (receives premium).

European: exercise only at T. American: exercise any time $\leq T$. (Assume European unless stated.)

Payoffs & Profits

Position	Payoff at T	Profit
Long call	$\max(S_T - K, 0)$	payoff $-Ce^{rT}$
Short call	$-\max(S_T - K, 0)$	payoff $+Ce^{rT}$
Long put	$\max(K - S_T, 0)$	payoff $-Pe^{rT}$
Short put	$-\max(K - S_T, 0)$	payoff $+Pe^{rT}$

C, P = option premium (time-0 price). Time-value of premium at T: Ce^{rT} or Pe^{rT}.

Bounds

Upper bounds: $C \leq S_0$ $\qquad P \leq Ke^{-rT}$

Lower bounds:

$$C \geq \max(0,\ S_0 - Ke^{-rT}) \qquad P \geq \max(0,\ Ke^{-rT} - S_0)$$

Bounds Arbitrage

$C > S_0$: short call, long stock.

	$t=0$	$S_T > K$	$S_T \leq K$
Short call	$+C$	$-(S_T - K)$	0
Long stock	$-S_0$	$+S_T$	$+S_T$
Total	$C - S_0 > 0$	$+K > 0$	$S_T \geq 0$

$P > Ke^{-rT}$: short put, invest Ke^{-rT}.

	$t=0$	$S_T < K$	$S_T \geq K$
Short put	$+P$	$-(K - S_T)$	0
Bond	$-Ke^{-rT}$	$+K$	$+K$
Total	$P - Ke^{-rT} > 0$	$S_T \geq 0$	$+K > 0$

$C < S_0 - Ke^{-rT}$: short stock, long call, invest $S_0 - C$ $[I \equiv (S_0 - C)e^{rT}]$.

	$t=0$	$S_T \geq K$	$S_T < K$
Short stock	$+S_0$	$-S_T$	$-S_T$
Long call	$-C$	$+(S_T - K)$	0
Invest	$-(S_0 - C)$	$+I$	$+I$
Total	0	$I - K > 0$	$I - S_T > 0$

$P < Ke^{-rT} - S_0$: long put, long stock, borrow $S_0 + P$ $[B \equiv (S_0 + P)e^{rT}]$.

	$t=0$	$S_T < K$	$S_T \geq K$
Long put	$-P$	$+(K - S_T)$	0
Long stock	$-S_0$	$+S_T$	$+S_T$
Borrow	$+(S_0 + P)$	$-B$	$-B$
Total	0	$K - B > 0$	$S_T - B \geq 0$

Put–Call Parity (European, no dividends):

$$C + Ke^{-rT} = P + S_0$$

		$t=T$ $S_T > K$	$S_T < K$
Port. A	Call	$S_T - K$	0
	ZC bond (K)	K	K
	Total	S_T	K
Port. C	Put	0	$K - S_T$
	Share	S_T	S_T
	Total	S_T	K

Same payoff $\Rightarrow PV_A = PV_C$: $C + dK = P + S_0$.

Put–Call Parity (General Form)

Let d = discount factor. Discrete: $d = (1+r)^{-T}$. Continuous: $d = e^{-rT}$.

$$C + dK = P + S_0$$

PCP Arbitrage (if parity violated):

$PV_A < PV_C$ ($C + dK < P + S_0$): Port. A cheap — buy call, short put, short stock, invest $S_0 + P - C$.

	$t=0$	$S_T > K$	$S_T \leq K$
Long call	$-C$	$+(S_T - K)$	0
Short put	$+P$	0	$-(K - S_T)$
Short stock	$+S_0$	$-S_T$	$-S_T$
Invest $S_0 + P - C$	$-(S_0 + P - C)$	$+Ie^{rT}$	$+Ie^{rT}$
Total	0	$Ie^{rT} - K > 0$	$Ie^{rT} - K > 0$

where $I = S_0 + P - C > dK$, so $Ie^{rT} > K$.

$PV_A > PV_C$ ($C + dK > P + S_0$): Port. C cheap — short call, buy put, buy stock, borrow $S_0 + P - C$.

	$t=0$	$S_T > K$	$S_T \leq K$
Short call	$+C$	$-(S_T - K)$	0
Long put	$-P$	0	$+(K - S_T)$
Long stock	$-S_0$	$+S_T$	$+S_T$
Borrow $S_0 + P - C$	$+(S_0 + P - C)$	$-Ie^{rT}$	$-Ie^{rT}$
Total	0	$K - Ie^{rT} > 0$	$K - Ie^{rT} > 0$

where $I = S_0 + P - C < dK$, so $Ie^{rT} < K$. Net profit $= K - Ie^{rT} > 0$ in all cases.

Trading Strategies

Bull spread: Long C_{K_1} + Short C_{K_2}, $K_1 < K_2$. Capitalise on price increase, cap max profit to reduce cost & risk.

S_T range	Long C_{K_1}	Short C_{K_2}	Total
$S_T \leq K_1$	0	0	0
$K_1 < S_T < K_2$	$S_T - K_1$	0	$S_T - K_1$
$S_T \geq K_2$	$S_T - K_1$	$K_2 - S_T$	$K_2 - K_1$

Bear spread: Short C_{K_1} + Long C_{K_2} *or* Long P_{K_2} + Short P_{K_1}. Profit if $S_T \downarrow$.

Butterfly ($K_1 < K_2 < K_3$, $2K_2 = K_1 + K_3$): Long C_{K_1}, Short $2C_{K_2}$, Long C_{K_3}. Low-vol bet ($S_T \approx K_2$).

S_T range	C_{K_1}	C_{K_3}	$-2C_{K_2}$	Total
$S_T \leq K_1$	0	0	0	0
$K_1 < S_T \leq K_2$	$S_T - K_1$	0	0	$S_T -$
$K_2 < S_T < K_3$	$S_T - K_1$	0	$-2(S_T - K_2)$	$K_3 -$
$S_T \geq K_3$	$S_T - K_1$	$S_T - K_3$	$-2(S_T - K_2)$	0

Straddle: Long call + Long put (same K, T). Profit if $|S_T - K|$ large.

Covered call: Long stock + Short call. **Protective put:** Long stock + Long put.

Binomial Option Pricing

Single period: $S \to uS$ or dS, risk-free rate r, $R = 1 + r$. No-arb requires $u > R > d$.

Risk-neutral probability: $q = \dfrac{R-d}{u-d}$, $1 - q = \dfrac{u-R}{u-d}$

$$C = \frac{1}{R}[q\,C_u + (1-q)\,C_d]$$

$C_u = \max(uS - K, 0)$, $C_d = \max(dS - K, 0)$ for call; $\max(K - \cdot, 0)$ for put.

Multi-period: backward induction. Each node same formula. **CRR:** $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$.

Replication Portfolio (Binomial)

Construct portfolio: x shares + b in risk-free cash. Payoffs:

$$\text{up: } ux + Rb = C_u \qquad \text{down: } dx + Rb = C_d$$

Solve:

$$x = \frac{C_u - C_d}{u - d} \qquad b = \frac{uC_d - dC_u}{R(u-d)}$$

$$C = x + b = \frac{qC_u + (1-q)C_d}{R} \quad \checkmark$$

$C < x + b$: Short x shares, borrow b, buy call. Net at $t=0$: $(x+b) - C > 0$. At T: portfolio payoff $-C_u$ or $-C_d$ cancels call payoff, leftover $= R((x+b) - C) > 0$.

$C > x + b$: Short call, long x shares, invest b. Net at $t=0$: $C - (x+b) > 0$. At T: call payoff $-C_u$ or $-C_d$ cancels portfolio, leftover $= R(C - (x+b)) > 0$.